

Linux Server Security Assignment - Questions and Answers

Question 1: Enable and configure SSH on your Linux server so that only key-based authentication is allowed and password authentication is disabled.

Answer:

Install OpenSSH Server:

```
sudo apt update  
sudo apt install openssh-server -y
```

Generate SSH Key:

```
ssh-keygen -t rsa -b 4096
```

Add Public Key:

```
cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_keys  
chmod 700 ~/.ssh  
chmod 600 ~/.ssh/authorized_keys
```

Edit SSH Configuration:

```
sudo nano /etc/ssh/sshd_config
```

Change:

```
PasswordAuthentication no  
PubkeyAuthentication yes
```

Restart SSH:

```
sudo systemctl restart sshd  
getenforce
```

Question 2: Set up a basic firewall using UFW or firewalld to allow only SSH (22) and HTTP (80) traffic and block all other incoming connections.

Answer (UFW):

```
sudo apt install ufw -y  
sudo ufw default deny incoming  
sudo ufw allow 22/tcp  
sudo ufw allow 80/tcp  
sudo ufw enable  
sudo ufw status numbered
```

Verify Open Ports from Another Machine:

```
nmap SERVER_IP
```

Expected Result:

```
22/tcp open ssh  
80/tcp open http
```

Question 3: Write a shell script automate_backup.sh that creates a backup of your /home directory and saves it with a timestamp in /backups.

Answer:

```
#!/bin/bash  
DATE=$(date +%F_%H-%M-%S)
```

```
mkdir -p /backups
tar -czvf /backups/home_backup_$(date +%Y%m%d).tar.gz /home
```

Make Executable:
chmod +x automate_backup.sh

Run Script:
./automate_backup.sh

Question 4: Check the current status of SELinux on your server, switch it to enforcing mode if it is not already, and verify the change.

Answer:
Check Status:
getenforce
sestatus

Enable Enforcing Mode:
sudo setenforce 1

Verify:
getenforce

Expected Output:
Enforcing

Question 5: Generate a checklist of at least 5 best practices for securing a Linux server and compare it with your server configuration.

Answer:
Best Practices Checklist:
1. Use SSH Key Authentication.
2. Disable Password Authentication.
3. Enable Firewall Protection.
4. Keep System Updated Regularly.
5. Enable SELinux in Enforcing Mode.
6. Use Strong User Passwords.
7. Disable Unused Services.

Comparison With Current Server:
✓ SSH Key Authentication Configured
✓ Password Authentication Disabled
✓ Firewall Enabled (Ports 22 and 80 Only)
✓ SELinux Enforcing Mode Enabled
✗ Automatic Security Updates Not Configured
✗ Unused Services Not Audited Yet

Missing Steps:
1. Configure automatic updates.
2. Review and disable unnecessary services.
3. Implement regular backup monitoring.